



Microbial community / environmental DNA analysis with Chipster

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Schedule

Day 1: Tuesday 14.4. (9-13) **Amplicon data processing**

Day 2: Wednesday 15.4. (9-13) **Microbial community data analysis**

Tips for the course

- You can ask questions anytime
 - Use your microphone
 - Write your question in the course Google doc: bit.ly/eDNA2026
 - If you get stuck in the exercises, ask for help
- When you are not speaking, please mute your microphone
- Introductory round to test your mics: Please tell
 - Your name
 - Institute
 - If you have microbial community / eDNA data already and how have you analyzed it
 - What is the best thing about spring

What will I learn?

- Microbial community / environmental DNA analysis of amplicon sequencing data
 - Central concepts
 - Analysis steps
 - File formats
- Microbial 16S rRNA gene data is used in the exercises
 - We discuss also how to analyze data on other marker genes
- How to operate the Chipster software

Understanding data analysis - why?

- Bioinformaticians might not always be available when needed
- Biologists know their own experiments best
 - Potential batch effects etc
- Allows you to design experiments better → less money wasted
- Allows you to discuss more easily with bioinformaticians

Introduction to microbial community / eDNA analysis

Outline

- What questions does it answer
- How is it done
- What are the main steps

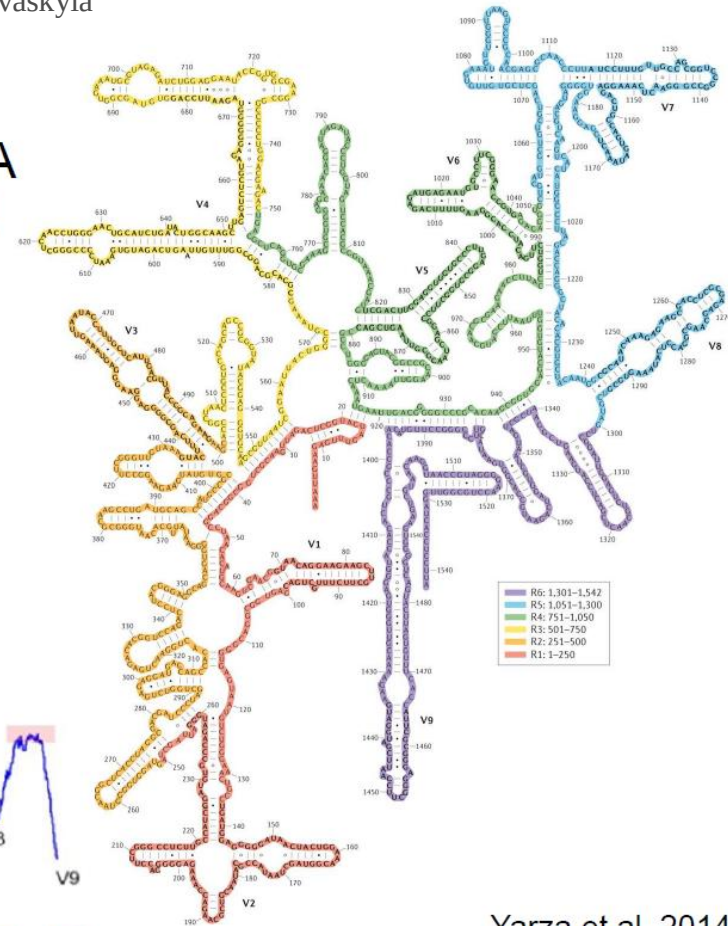
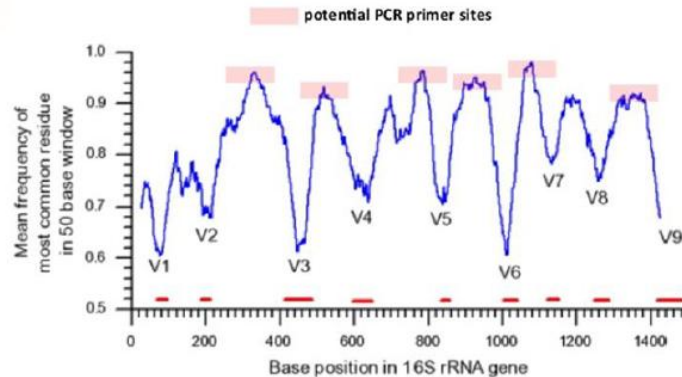
Microbial community / eDNA analysis

- Answers the questions **who are there** and **in what proportions** if compared to your other samples
 - It will not confirm that someone isn't there (sampling depth, primer/sequencing bias)
- Specific primers are used to amplify a region of one gene
 - Bacterial and archaeal communities: 16S rRNA
 - Fungal communities: ITS (internal transcribed spacer between 18S and 5.8S rRNA genes)
- Sequenced using Illumina MiSeq (or Ion Torrent)
 - Also long read sequencing with better resolution: PacBio, Oxford Nanopore
- Different from metagenomics, where the aim is to sequence all genes
 - Answers the questions **who are there** and **what are they capable of doing**

Commonly used 16S rRNA gene amplicons are called by the variable regions they contain

- V1-V2
- (V3)-V4

Variable Regions of the 16S rRNA:



Yarza et al. 2014

Nature Reviews | Microbiology

UNIVERSITY OF JYVÄSKYLÄ

Main parts of microbial community analysis



- **Preprocessing**

- Quality control, trim primers/adaptors and bad quality ends
- Filter and trim sequences based on quality scores and/or length

- **Processing reads into ASV/OTUs**

- Create contigs (paired end reads)
- Cluster reads into ASVs/OTUs
- Remove chimeras and sequencing errors
- Taxonomic assignment (e.g. SILVA for 16S, UNITE for ITS)

- **Community analysis and visualization**

- Does community structure differ between sample groups?
- Which taxa are differentially abundant between sample groups?

How to choose the preprocessing protocol?

- Sequencing technology:
 - Illumina Miseq: paired-end short reads
 - Ion Torrent: single-end short reads
 - PacBio and Nanopore: long reads
- Gene: 16S rRNA, ITS, other?
 - reference database
 - gene characteristics
- Operational taxonomic units (OTUs) vs. amplicon sequence variants (ASVs)
 - OTUs: mothur, QIIME2
 - ASVs: DADA2

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 - **ASVs: DADA2**

OTUs vs. ASVs

- Same purpose: taxonomic grouping and error correction
- ASV = amplicon sequence variant
 - after error correction, each sequence is a different ASV
 - different copies of 16S rRNA gene from the same organism are different ASVs
- OTU = operational taxonomic unit
 - sequences clustered into 'species' based on a similarity cut off
 - bacterial 16S rRNA gene → 97 %
 - very closely related species based on short reads cluster into the same OTU

→ split or clump?

Processing data of other genes than 16S, 18S or ITS

- Similar pre-processing workflows can be followed as with 16S, 18S or ITS data, but you have to provide the following files:
 - OTU workflow: Mothur-formatted reference sequence FASTA file and taxonomy file
 - ASV workflow: DADA2-formatted reference sequence FASTA file
- Depending on the gene, you might have to skip the alignment step of the OTU workflow or align the sequences outside Chipster.

OTU-based analysis with mothur in Chipster

- Pre-processing: Microbial amplicon data preprocessing for OTU
 - Remove primers and adapters
 - Pair reads into contigs
 - Remove identical sequences
 - Alignment & filter and trim alignment
 - Precluster
 - Remove chimeras
 - Classify sequences (= assign taxonomy)
 - Cluster sequences into OTUs & classify OTUs
- ➡ Check videos on the Chipster Youtube channel for details
- ➡ All mothur materials:
<https://chipster.2.rahtiapp.fi/manual/courses.html#microbiome>

Fungal ITS & DADA2

Best practices in metabarcoding of fungi: From experimental design to results

[Leho Tedersoo](#) ✉ [Mohammad Bahram](#), [Lucie Zinger](#), [R. Henrik Nilsson](#), [Peter G. Kennedy](#), [Teng Yang](#), [Sten Anslan](#), [Vladimir Mikryukov](#)

First published: 08 April 2022 | <https://doi.org/10.1111/mec.16460> | [VIEW METRICS](#)

Glomeromycota and nonfungal eukaryotes (Figure Box 2). In terms of community composition, the results are similar between ESV and OTU-based approaches (Glassman & Martiny, 2018; Porter & Hajibabaei, 2020), because these are driven by abundant taxa. We conclude that ESV approaches overestimate richness of common fungal species (due to haplotype variation) but underestimate richness of rare species (by removing rare variants; see also Joos et al. 2020). ESV approaches can nevertheless be useful for studying allele or haplotype distribution of various common species based on eDNA (Zizka et al., 2020).

→ consider OTU-based approach with ITS



Fungal Ecology
Volume 65, October 2023, 101274



Mini-review

ITS alchemy: On the use of ITS as a DNA marker in fungal ecology

[Håvard Kausrud](#) ✉

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Since numerous ITS alleles often occur within fungal species, it is clearly not suitable to use so-called amplicon sequence variants (ASVs) as units in ITS based fungal community studies. For markers with a high level of intraspecific variation, the diversity will be tremendously overestimated by treating each ITS haplotype as a biological entity in downstream statistical analyses (Fig. 2b). Due to its high sensitivity, it has also been shown that DADA2 splits single bacterial genomes into several ASVs because of intragenomic variability, which is another concern also relevant for the fungal ITS marker (see below). The ASV term is connected to the bioinformatics program DADA2 (Callahan

Data set for exercises: mouse fecal samples



- Subset of 19 mouse fecal samples
- Two time points:
 - early: collected on days 0-9 post weaning
 - late: collected on days 141-150 post weaning

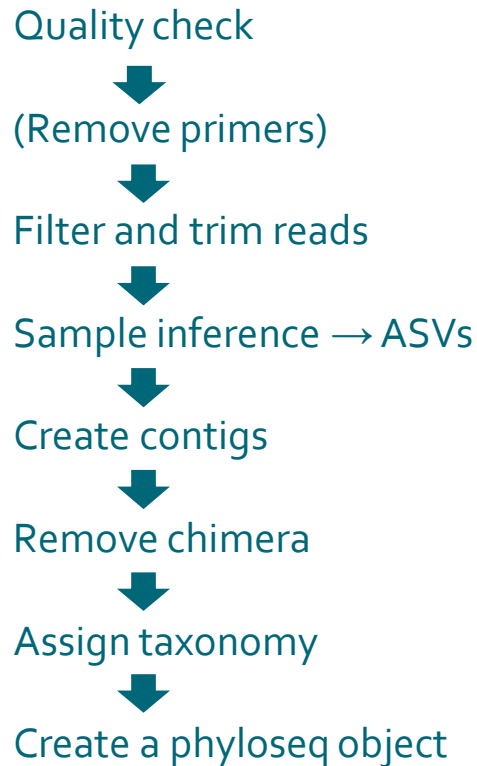
→ Does mouse gut microbiome change?

Data set for exercises: mouse fecal samples

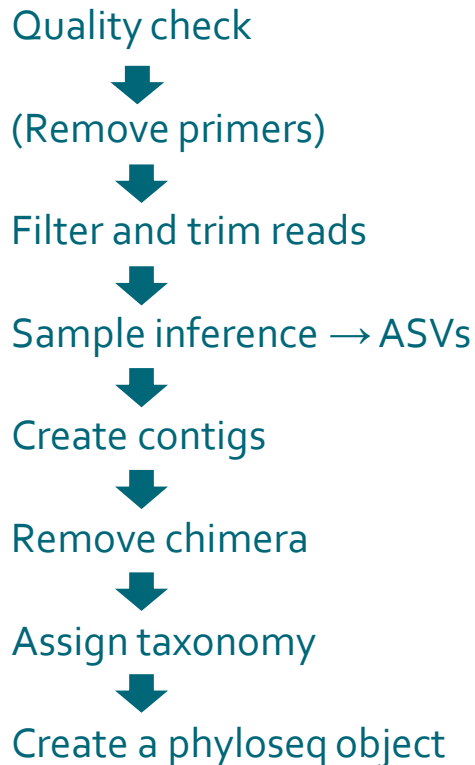


- PCR amplification of V₄ region of the bacterial 16S rRNA gene (ca. 250 bp)
- Sequencing by Illumina MiSeq
- Demultiplexed (separated by sample) and barcodes removed
- Primers and sequencing adapters removed
- Each sample in two separate FASTQ files:
 - forward reads (R1)
 - reverse reads (R2)

Outline of Day 1:



Outline of Day 1:



Output so far: phyloseq object with

1. ASV table (counts in each sample)
2. taxonomy of ASVs
3. sample information (phenodata)
4. sequences of ASVs

Day 2:

- Data tidying & transformations
- Taxonomy plots
- Alpha diversity
- Beta diversity: ordinations & statistics
- Differential abundance analysis